

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 11

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive (1) • Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected (1) • Damage to other cell parts can stop the cell functioning or kill it; low damage may be repaired by the intact nucleus (1)	3
01.2	<i>1 mark for each point.</i> • Acute: effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) (1) • Chronic: effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) (1)	2
01.3	<i>1 mark for each point.</i> • Exposure: absorption of ionising radiation from a source the person is close enough to (1) • Contamination: physically carrying radioactive material on or in the body, so exposure continues wherever the person goes; sealed sources can expose but not contaminate (1)	2
01.4	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Sieverts = grays x radiation weighting factor (20 for alpha) (1) • $0.002 \times 20 = 0.04 \text{ Sv}$ (40 mSv) (1)	2
01.5	<i>1 mark.</i> • The activity of a source (number of decays per second) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
02.1	1 mark for the correct answer. • $(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%) (1)	1
02.2	1 mark for each point. • A fixed fee gives no incentive to save water because the bill is the same however much is used (1) • Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) (1)	2
02.3	Any 3 from the points below - 1 mark each, maximum 3. • Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers (1) • Greywater re-use: bath / shower water used to flush toilets or water gardens (1) • Rainwater collection / water butts for garden use (1) • Xeriscaping: planting drought-adapted species that need no watering (1) • Repairing leaks / dripping taps (1)	3
02.4	1 mark for each point. • Old infrastructure / pipes in poor condition leak (1) • Repair and good maintenance / pipe replacement / leak detection reduce wastage (1)	2
02.5	1 mark for each point. • Water is delivered directly to the plant roots / soil around each plant (1) • Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
03.1	Correct answer only. • Organic (methyl) mercury (1)	1
03.2	1 mark for each point. • Anaerobic microbes converted the inorganic mercury into organic methyl mercury (1) • Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people (1)	2
03.3	1 mark for each point. • Most heavy metals are less soluble at high pH (1) • So they remain immobile and are not leached into water or absorbed by organisms (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier (1) • Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground (1) • Take repeated readings at each distance and calculate means (and standard deviation) to assess variability (1) • Use a measure such as LA10 or Leq over a set period rather than a single peak reading (1) • Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance (1)	5
04.2	<i>1 mark.</i> • Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier (1)	1
04.3	<i>1 mark for each point.</i> • LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods (1) • LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level (1)	2
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Double / triple glazing (1) • Acoustic insulation in walls / roof (1) • Bedrooms on the side facing away from the track (1) • Dense hedges / trees and earth mounds in gardens (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
05.1	<i>1 mark for each point.</i> • Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) (1) • Magnetometry (1) • Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata (1) • Resistivity (1)	4
05.2	<i>1 mark for each point.</i> • Trial drilling is the most expensive technique per site (1) • Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites (1) • Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) (1)	3
05.3	<i>1 mark for each point.</i> • Different minerals emit / reflect infrared at different wavelengths (1) • Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition (1)	2

Question	Marking guidance	Marks
05.4	1 mark. Results are available immediately in the field, saving time waiting for laboratory analysis (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
06.1	1 mark for the correct answer. 7×10^{11} tonnes (1)	1
06.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. 10% of 10 000 = 1000 billion tonnes (1) $1000 / 700 \times 100 = 143\%$ (accept 142.9) (1)	2
06.3	1 mark for each point. - Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) (1) - Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion (1)	2
06.4	1 mark for each point. - CO₂ dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis (1) - Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons (1) - When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods (1)	3
06.5	1 mark for each point. - Aerobic respiration uses oxygen to break down organic compounds completely to CO₂ (and water), releasing much more energy (1) - Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH₄), which is later oxidised to CO₂ in the atmosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
07.1	1 mark for each point. - Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines (1) - Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost (1)	2
07.2	1 mark for each point. - As volume increases the surface area : volume ratio decreases (1) - Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage) (1)	2
07.3	1 mark for each point. - Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation (1) - Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • Materials with high specific heat capacity (concrete, water) warm and cool slowly (1) • They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating (1)	2
07.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation (1) • Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage (1) • Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	
08.1	<i>1 mark for each point.</i> • Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable (1) • Wild species have different ranges of tolerance so cross results cannot be applied directly (1)	2
08.2	<i>1 mark for each point.</i> • Preliminary study with widely spaced pH values to find where germination occurs (1) • Final range from below the lowest pH giving germination to above the highest, with closer intervals (1)	2
08.3	<i>1 mark for each point.</i> • Temperature (constant, natural germination temperature) (1) • Light, water volume, number of seeds per dish, seed source (1) • Growth medium (filter paper; soil would alter pH); nutrients not needed for germination (1) • Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure (1)	4
08.4	<i>1 mark for each point.</i> • Surface water affects fresh mass and drying damages the plants; dry mass requires killing the plants so the same plants cannot be re-measured (1) • Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff (2) • Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover (2)	4

Question	Marking guidance	Marks
09.2	1 mark for each point. • Tree roots that held the soil together decompose so the soil is bound less strongly (1) • Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely (1)	2
09.3	Any 4 from the points below - 1 mark each, maximum 4. • Sedimentation in rivers reduces flow capacity so flooding downstream is more likely (1) • Sedimentation in reservoirs reduces their storage volume (1) • Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants (1) • Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas (1) • Desertification as vegetation cannot re-establish; reduced rainfall (1) • Nutrients (phosphates) carried on soil particles causing eutrophication (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	
10.2	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Biological control (introduced predators / pathogens) (1) • Predator habitats (hedgerows, beetle banks) (1) • Crop rotation (1) • Companion / barrier crops (1) • Sterile male technique / pheromone traps (1) • Genetic resistance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25

Question	Marking guidance	Marks
11.2	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25